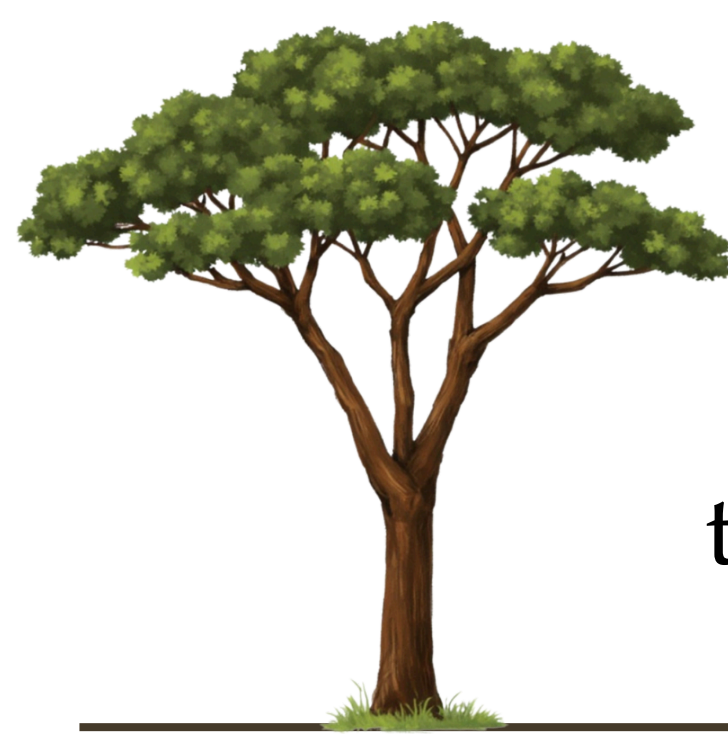




Tree of Thoughts (ToT)

teaching language model to plan, explore, and backtrack

Jane Tenecota-Villa, Jade Lee, Lauren Ah-Hot

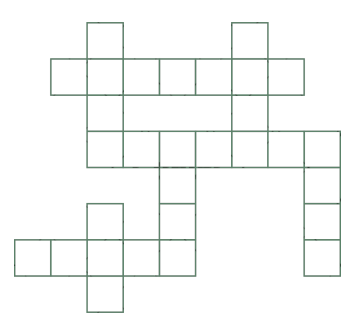
based off of Yao et al., NeurIPS 2023

Cornell University
CS 4782 Deep Learning

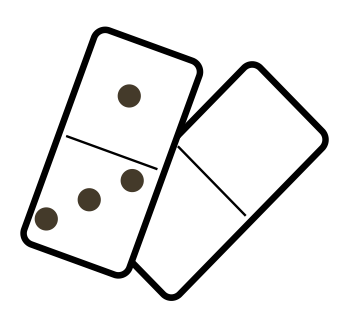


Motivation

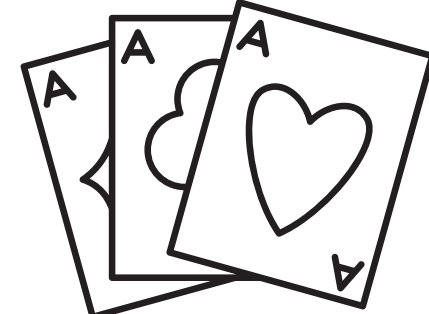
LLMs can *fail* real-puzzles and games that represent real world problems



evaluating partial solutions



requires exploring possibilities



revising earlier decisions

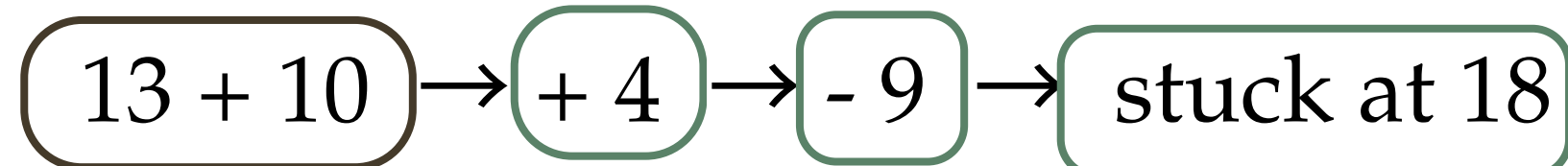
CAUSE of problem: current IO(input/output)/CoT (chain of thought) prompting cause LLMs to reason in a straight line (left to right), meaning they cannot revise early mistakes or explore alternatives.

THE GAME OF 24

4 9 10 13

use each number once with $+$ $-$ $\times$ $\div$ to make 24

Paper Results on direct prompting: IO (~7%) | CoT (~4%)



Thus, ToT reframes reasoning as search: it generates candidate steps, scores them, and backtracks to explore better paths.

GOAL: Test whether ToT (search-based reasoning) outperforms standard prompting (IO, CoT) on the Game of 24
(paper says yes: ToT ~74%)

References

[1] Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2023). *Tree of Thoughts: Deliberate Problem Solving with Large Language Models.* In NeurIPS 2023. arXiv:2305.10601.

[2] OpenAI (2024). GPT-4o. <https://openai.com/index/hello-gpt-4o/>

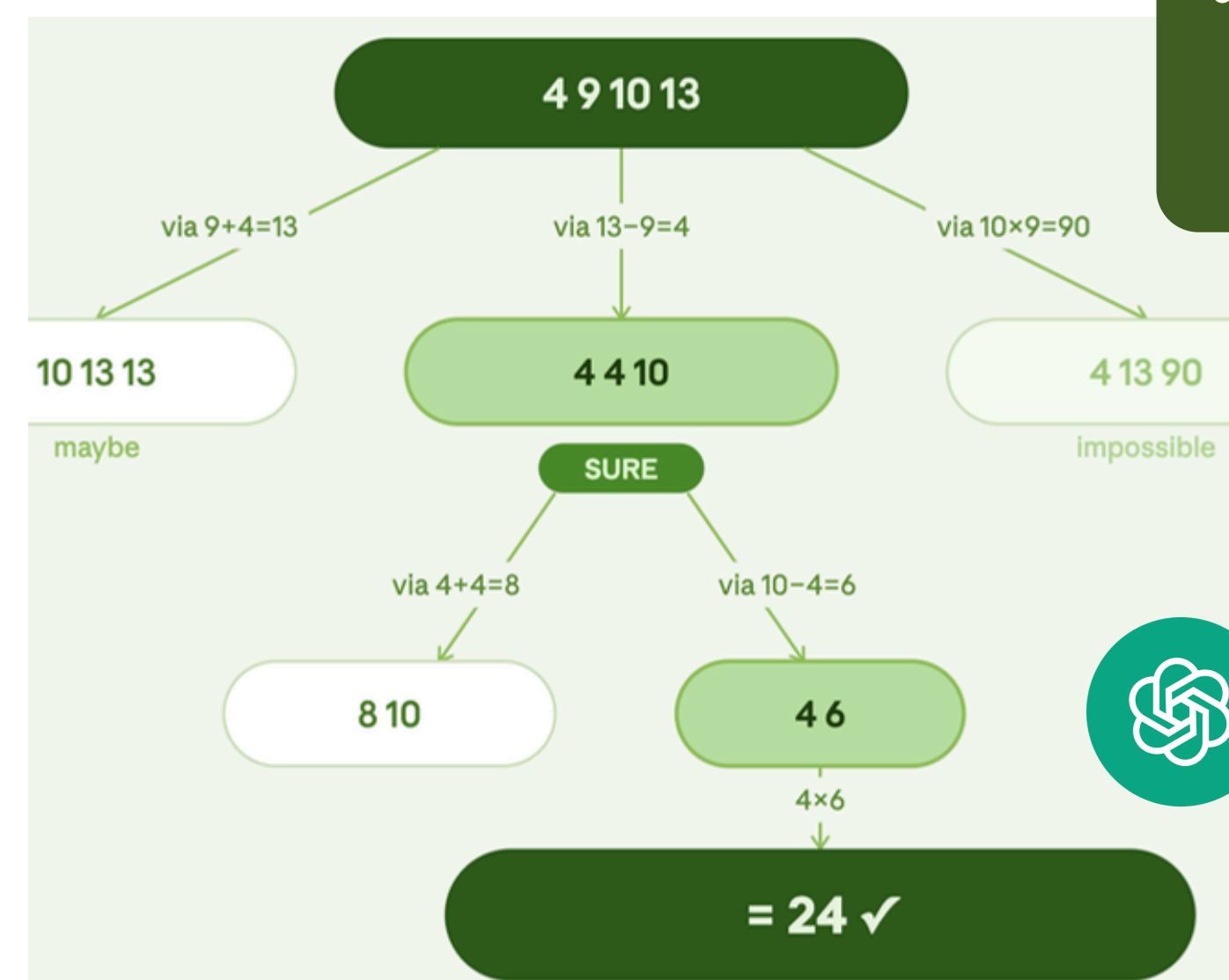
[3] 4nums.com Game of 24 dataset. github.com/princeton-nlp/tree-of-thought-llm

Methodology

- **Task:** Game of 24
- **Dataset:** 50 puzzles randomly sampled from the official 4nums.com dataset- paper used
- **Model:** GPT-4o via the OpenAI API

Modifications

- 50 problems instead of 100
 - (budget constraint)
- GPT-4o instead of GPT-4 (outdated was used in May 2023, and cheaper)
- Few-shot evaluator with our own examples (paper prompts reused for IO/CoT)

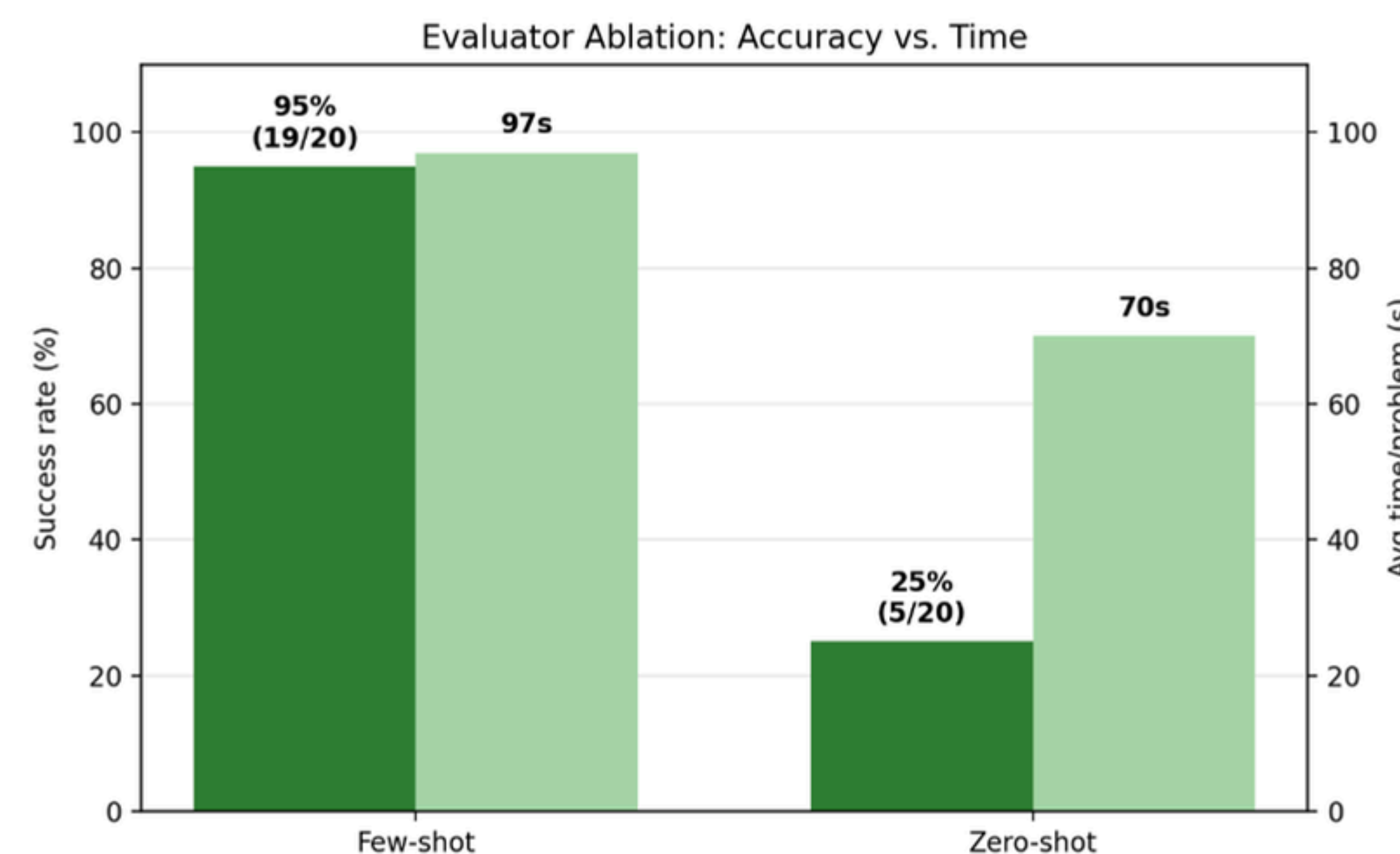


Three Prompting Conditions

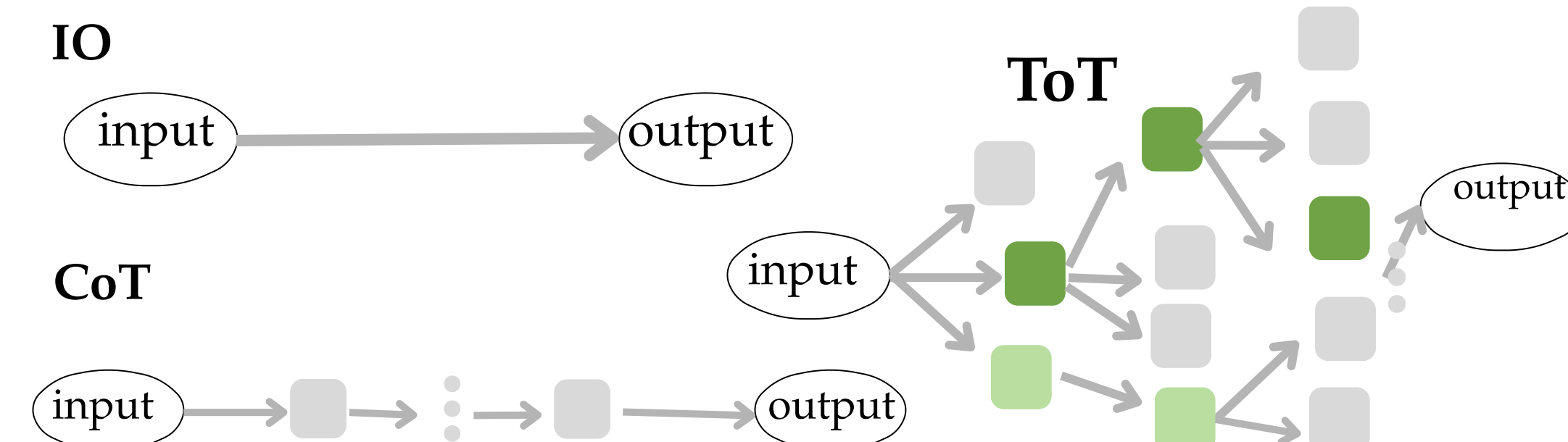
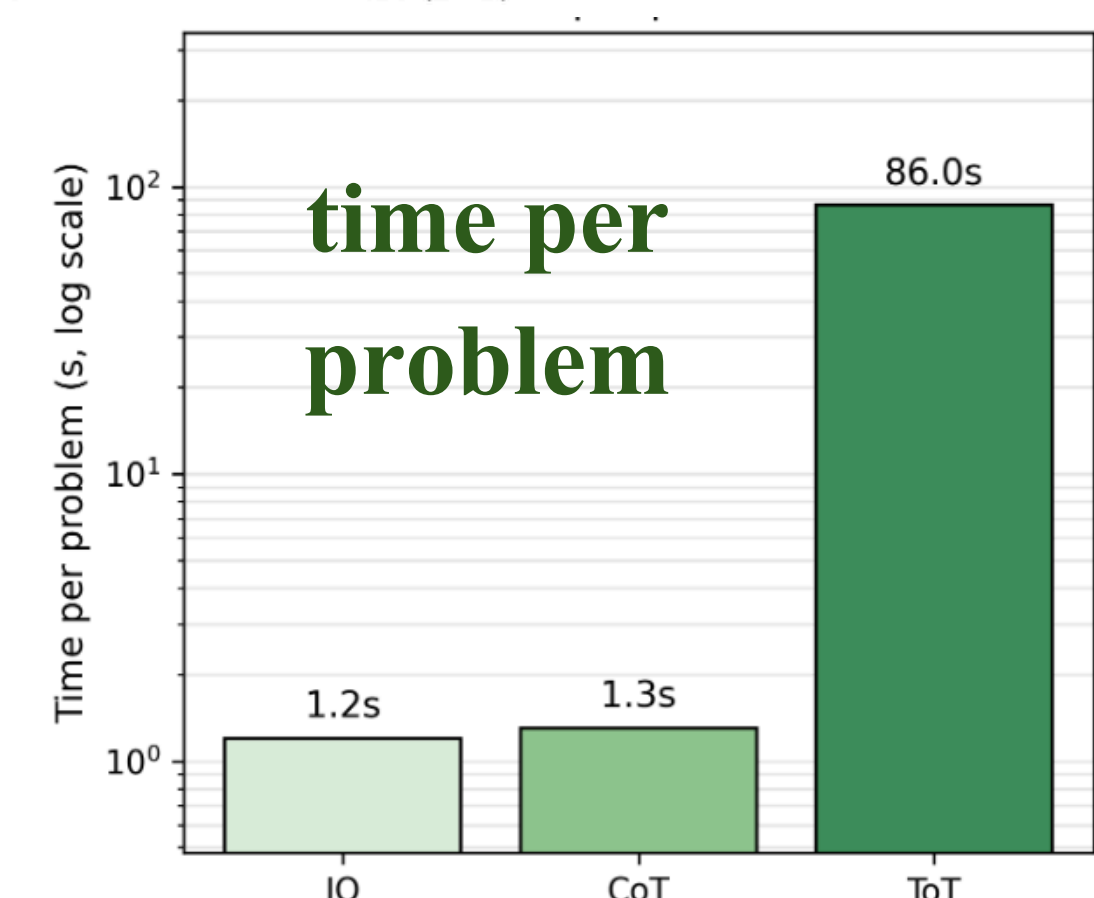
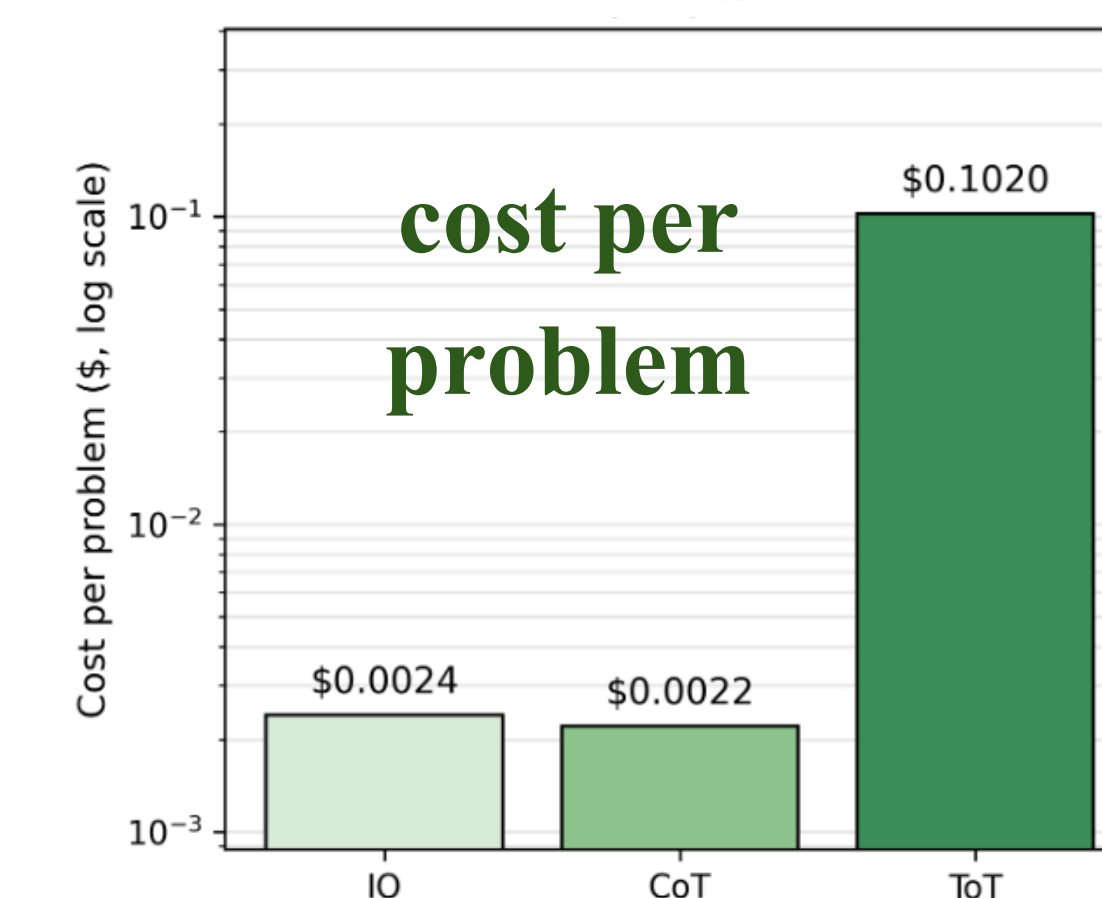
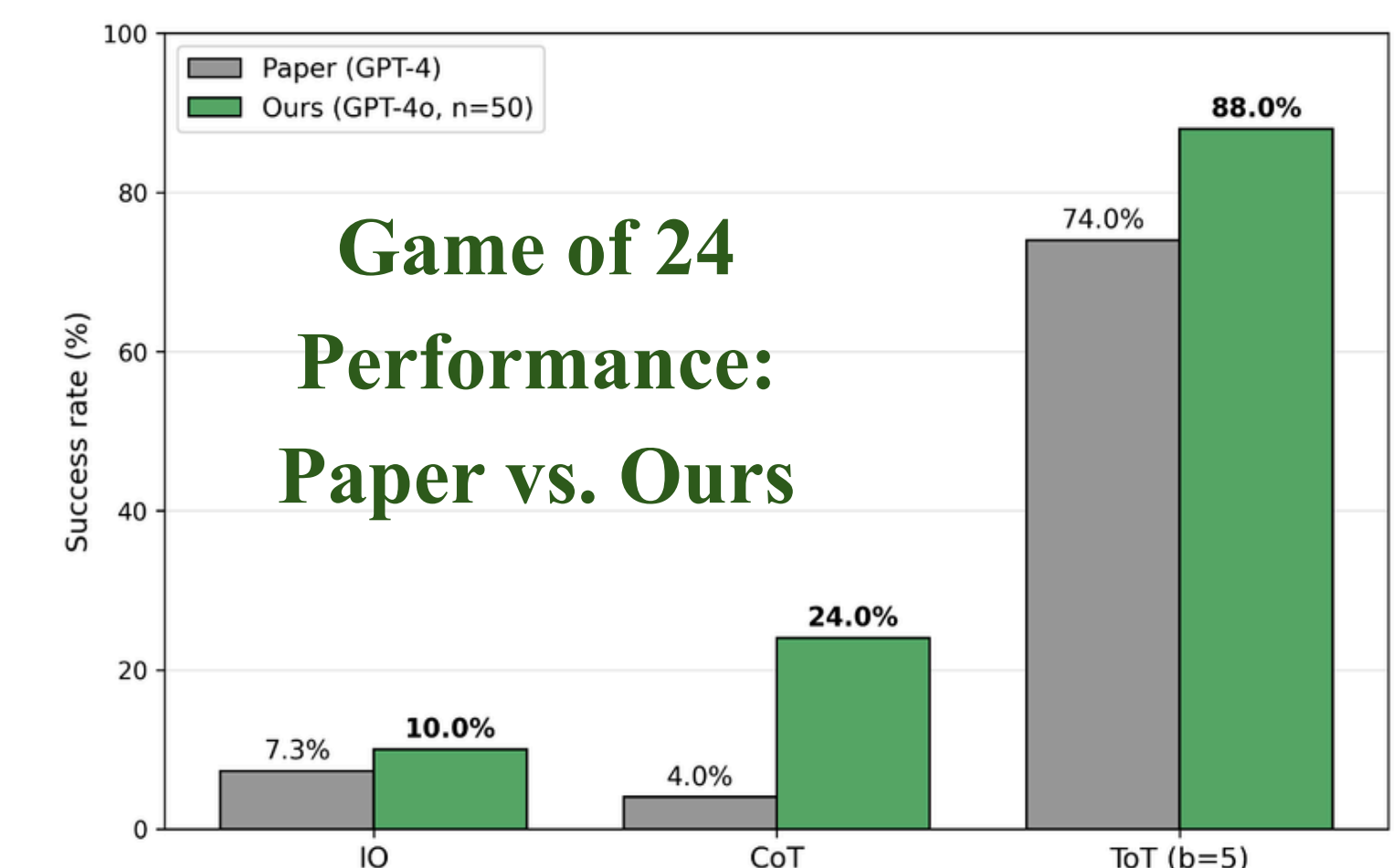
- **IO:** 5 shot, direct answer only
- **CoT:** 5 shot, step by step reasoning along answer
- **ToT (BFS):**
 - 3 reasoning steps (depth = 3)
 - generate candidate next steps
 - Evaluator: score each step
 - (sure / maybe / impossible)
 - Search: keep top 5 (beam size = 5)

Further exploration

Comparing evaluators: Same 20 puzzles, same proposer, same beam, only the evaluator prompt changes (zero-shot vs. our 7-example few-shot)



Results & Evaluation



Conclusion & Future Work

- **Replication holds:** ToT > CoT & IO on Game of 24 (88% vs. 24% vs. 10%)
- **Few-shot evaluator is critical:** removing it drops 95% → 25%
- **Cost tradeoff:** ToT ~45× more expensive per problem
- **But total cost is low:** ~\$5 for 50 problems (with GPT-4o)
- **Cost per correct is closer:** IO/CoT waste spend on failures
- **Future work (DFS):** prune early to cut cost, keep ~88% accuracy